

Marco Liu

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EDUCATION

University of California, San Diego <i>San Diego, CA</i>	<i>09/2023 - 03/2025</i>
<i>Master of Science in Computer Science GPA: 3.86/4.0</i>	
The Chinese University of Hong Kong <i>Hong Kong</i>	<i>09/2019 - 07/2023</i>
<i>Bachelor of Science in Computer Science GPA: 3.73/4.0</i>	

SKILLS

Programming Languages: Java, Python, JavaScript, TypeScript, C, C++, Go, SQL, HTML, CSS

Web Technologies: Node.js, Express, React, Next.js, Redux, Nest.js, Spring Boot, MySQL, PostgreSQL, MongoDB, Prisma, TypeORM, MyBatis, PrimeReact, Tailwind CSS, Swagger/OpenAPI, Jest, JUnit, Mockito

Others: AWS(EC2, ECS, Lambda, IAM, RDS), Azure, GCP, OpenAI, Docker, Redis, Elasticsearch, Kafka, Linux, Git, Figma, CI/CD, Agile

EXPERIENCE

curaJOY - <i>Family Wellness Solutions Non-profit</i>	<i>05/2024 - Present</i>
Software Developer	<i>San Diego, CA</i>

- Developed an inclusive, gamified, and AI-assisted platform for enhancing family behavioral wellness, building 10+ reusable UI components with **React**, **Next.js**, **TypeScript**, and **PrimeReact**, while enhancing UI with **Tailwind CSS**.
- Enhanced **user-interactive 3D avatars** with **Three.js**, incorporating animations, advanced visuals, and seamless integration with a sophisticated behavior, mood, and body check-in system, boosting user satisfaction by 40%.
- Built and optimized reward and check-in systems with **Redux** modules, handling asynchronous data fetching, complex state updates, and user interaction logic, resulting in a 20% improvement in app performance and increased retention rates.
- Operated in an **Agile(Scrum)** environment using **Jira** to manage sprints, accelerating development cycles and delivering new features and releases more frequently through improved team coordination and project management.

Orbiz.app - <i>Influencer Connection AI SaaS Startup</i>	<i>04/2024 - Present</i>
Software Engineer Intern	<i>San Diego, CA</i>

- Developed an **LLM-powered**, data-driven platform and a **Chrome extension** to streamline influencer outreach and campaign creation, reducing manual effort by 15 hours/week and improving marketing campaign efficiency.
- Designed and implemented scalable database schemas using **PostgreSQL** and **TypeORM**, leveraging hierarchical relationships and entity mappings to optimize complex workflows for campaign creation and influencer matching.
- Integrated APIs with **Azure OpenAI** to scrape websites, analyze products to generate campaign suggestions and relevant tags.
- Designed and deployed **REST APIs** with **Node.js**, **Nest.js**, **Express** and **TypeScript**, enabling efficient influencer querying, exporting, and tag-based matching across 3,000,000+ records in under 4 seconds using optimized SQL queries.
- Orchestrated dual **AWS RDS** connections with **Docker** containers, managed **CI/CD** pipelines via **GitHub Actions** and **AWS CodePipeline** on **ECS**, and configured **IAM** task roles with execution policies for secure and dynamic access to AWS resources.

Chinese University of Hong Kong, KEEP - <i>Leading eLearning Platform for Higher Education Institutions</i>	<i>09/2022 - 04/2023</i>
Software Developer	<i>Hong Kong</i>

- Designed and developed a video analysis module for an **AI-powered** online interview assessment platform, integrating Microsoft **Azure** AI services and open-source **ML** frameworks such as **OpenCV** and **Google MediaPipe**, boosted analysis accuracy by 30%.
- Implemented algorithms in **Python** to automate interview scoring, provide insightful feedback, and detect cheating behaviors, achieved 93% accuracy in analysis and 40% reduction in processing time, decreased server costs by \$2,500.
- Utilized **React**, **TypeScript**, and **Material UI** to create interactive dashboards and real-time camera calibration, enhancing user interaction and dynamic data visualization, while employing **MySQL** and **Prisma** to manage 2,000+ video evaluations efficiently.

SELECTED PROJECT

VivaVend - <i>A Distributed E-Commerce Platform</i> <i>San Diego, CA</i>	<i>11/2023 - 02/2024</i>
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- Developed a high-availability, distributed E-Commerce platform using **Java/Spring Boot**, deployed on **AWS EC2**, configured microservices and load balancing with **Spring Cloud Gateway**, **Consul**, and **OpenFeign** for enhanced scalability.
- Optimized event handling with **Kafka**, supporting up to 100K QPS during peak times, reducing database load by 95%, and implemented distributed locks with **Redis** and **Lua** script for inventory caching to prevent overselling.
- Boosted product search speed by 90% using **Elasticsearch** with inverted index search, outperforming **MySQL** fuzzy queries.
- Implemented a subscription notification workflow by integrating **AWS Lambda**, **S3**, **EventBridge**, and **IAM**, enabled seamless database monitoring and real-time email notifications via Amazon **SNS**.